Unit 6 End-of-Unit Assessment - Form B Teacher Key

8 items · 3 points each · 24 total. Award 1 for a valid model/setup, 1 for correct mathematics/units, and 1 for a requested justification. Use the applicable evidence when the task is conceptual; accept equivalent methods and response modes. Do not count writing length as mathematical mastery.

Item 1

Prompt: How many 1/4-meter portions fit in 5 meters? Model and check.

Solution / evidence: 20; 20×1/4=5 m.

Item 2

Prompt: A 11/8-meter strip is divided by a 1/4-meter group size. Find the numerical quotient and complete-group count.

Solution / evidence: 11/2 = 5.5 groups numerically; 5 complete groups, with 1/8 m left.

Item 3

Prompt: Expand and evaluate 5^3. Explain why it differs from 5×3.

Solution / evidence: 5×5×5=125; 5×3=15.

Item 4

Prompt: Evaluate 30÷5×2 and $(2+5)^2$. State the order used.

Solution / evidence: 12 and 49; division/multiplication left to right; grouping then power.

Item 5

Prompt: Write a cost expression for $5 per hour plus one $7 fee. Define h.

Solution / evidence: 5h+7 dollars; h=hours.

Item 6

Prompt: Evaluate 5x+7 at x=4. Show substitution.

Solution / evidence: 27; 5(4)+7.

Item 7

Prompt: Compare 3(x+5) and 3x+15 using properties; compare 3x+15 and 6x using a counterexample.

Solution / evidence: First equivalent by distribution. Second fails at x=1: 18 versus 6.

Item 8

Prompt: Find GCF of 10 and 15, then factor 10x+15 with the GCF and expand to check.

Solution / evidence: GCF 5; 5(2x+3); expansion restores both terms.

Response: group errors by interpretation/model versus calculation versus explanation. Use the corresponding lesson Teaching Support Guide, revise one step, and reassess with its reserved fresh check. Do not reuse the already seen form as independent evidence.